

Modeling and Simulation of Three Phase Inverter for Fault Study of Microgrids

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Abstract—Single phase and three phase voltage source inverters are increasingly used to integrate electric power generation from solar photovoltaics with the electric distribution systems. The fault characteristics of an inverter interfaced generator is different from the fault characteristics of conventional synchronous generators because inverters are designed to restrict their fault currents to a low value. Such restriction in fault current thus poses a challenge to conventional protection schemes. In order to explore fault studies of distribution systems with inverter-interfaced generation, and to design appropriate protection schemes for such systems, a suitable inverter model is required. The purpose of this paper is to model and simulate a three phase inverter that is able to perform well in the presence of balanced or unbalanced loads, and is also able to restrict the fault currents effectively without significant distortions of voltage and current waveforms.

I. INTRODUCTION

Electric power generation from renewable sources is ever increasing due to several reasons including energy security and environmental concerns. Solar photovoltaic is one of the major contributors to renewable power generation. Output voltage and current from solar photovoltaics are direct current (dc). Hence power electronic inverters are used as an interface while connecting these sources to alternating current (ac) distribution systems. These inverters are actively current limited, and have small fault current contribution ($\approx 1 - 2$ pu) due to low thermal inertia of power electronic switches, as opposed to conventional generators ($\approx 5 - 10$ pu) [1], [2]. This limited fault current capability of inverters may lead to protection problems, because over-current based conventional protection schemes can not be employed to detect such low fault currents.

Synchronous generators conventionally used in power systems generate only positive sequence voltages while supplying load or fault. They also have high fault current capability. In contrast, in order to protect the component devices, inverters are required to limit their fault current capability. This is accomplished through reduction in the internal voltage during faults. Phase voltages of three phase inverters during fault may comprise of (1) only positive sequence component, or (2) positive and negative sequence components, or (3) positive, negative and zero sequence components depending upon inverter configuration and fault type [3].

Fault behavior of an inverter is determined by its control, which in turn is based on the inverter topology (three phase

three wire, three phase four wire, single phase) and topology of the network to which the inverter is connected (balanced or unbalanced, and stiff or weak). Different control techniques for single phase and three phase inverters have been discussed in [4]–[10]. These control techniques can be broadly divided into the following three categories:

- 1) control in synchronously rotating $dq0$ reference frame,
- 2) control in stationary $\alpha\beta0$ reference frame,
- 3) control in natural reference frame.

Control in rotating reference frame is the most popular approach, where measured output voltages and currents are transformed into their equivalents in rotating frame ($dq0$) of reference [5]. These components would appear as dc quantities for balanced operation of three phase inverters. Simply proportional and integral (PI) controllers would suffice for satisfactory operation of inverters with control implemented in $dq0$ frame of reference. For unbalanced operation however, additional control measures need to be taken. Two different measures reported in literature are three phase control in sequence domain [7], [11], and per-phase control [8]. The $dq0$ based control technique used in three phase inverters can be extended to the dq control of single phase inverters [4], [10]. Moreover, based upon the control of single phase inverters in dq frame of reference, individual phase control of three phase inverters is accomplished as reported in [8] for supplying unbalanced loads.

Control of three phase inverters in stationary reference frame involves transformation of measured voltages and currents into stationary $\alpha\beta0$ quantities. Then the control is implemented in $\alpha\beta0$ frames of reference using either proportional resonant (PR) controller, or sinusoidal command following. The later approach is good only for constant frequency operation of an inverter (stand alone operation of an inverter) [5]. The details of PR controllers have been discussed in [12].

Controls in natural reference frame is not significantly different from control in stationary reference frame. The only difference is that the control in natural reference is implemented in phase domain rather than in stationary reference frames [13].

Different inverter topologies are investigated using the three categories of control techniques described in the previous paragraphs. Control of three phase four leg inverters (they have an extra pair of switches forming the fourth leg) in $dq0$

frame of reference has been discussed in [7]. 3-D space vector modulating (SVM) technique was implemented to generate gating signals for the IGBT bridge used in the device. Dai *et al.* [9] have proposed a control technique for three phase four wire inverter which has an added advantage during low dc bus voltage. This algorithm is a combination of discrete time sliding mode control and robust servomechanism control. The control algorithm was implemented in stationary frame of reference, and modified SVM technique was used to generate gating signals. Ninad and Lopes [8] have simulated a three phase three wire inverter operating in stand alone mode. Per phase based dq control was implemented in order to supply unbalanced loads. Baggu and Chowdhury [11] have demonstrated an implementation of a three phase three wire inverter in sequence domain to counter voltage unbalance. Voltage and current signals were first transformed into positive and negative sequences. Then rotating reference frame based PI controllers were implemented for each sequence component. The output signals from positive and negative sequence controllers were summed together to obtain modulating signals for different phases. Control of single phase inverters using d and q axes controllers is discussed in [4], [6]. Jung and Sul [10] have demonstrated an implementation of a single phase grid connected inverter using FPGA. Proportional-Resonant (PR) controller was used to regulate output current of the inverter. Plet *et al.* [13] have presented a comparative study of inverters' performance controlled in two different reference frames: synchronous reference frame, and natural reference frame. Three phase four wire inverter, operating in standalone mode to supply three phase load through a low voltage line, was simulated. Simulation results obtained with synchronous reference frame control were inferior than those obtained with natural reference frame control during fault. This difference is obvious, because $dq0$ based three phase control distorts the phase voltages and currents when there is unbalanced faults applied to the inverter system. Also, $dq0$ frame based control is not suitable in environments where loads may be significantly unbalanced.

Three phase PV inverters mostly use three phase four wire topology rather than three phase three wire [14]. However, three phase three wire inverter topology is very popular in the research community, and it is extensively used to study inverter behavior [15]–[19].

The purpose of this paper is to model and test an inverter that gives good performance during healthy and faulted conditions. The model will be used for future work involving protection and fault analysis of microgrids. Quality of the voltage and current waveforms at the output of the inverter is chosen as the main performance criteria. The waveforms must be as close to sinusoidal as possible, and the current output during fault must be limited very efficiently to a pre-determined value. Since this inverter model is to be used with distribution systems, the load model is formulated with high unbalance. As the literature survey has shown that per phase based dq control gives superior performance under unbalanced condition, this control methodology is adopted for the pro-

posed model. Using this methodology, the performances of the proposed model using three phase three wire and three phase four wire inverter topologies are compared. Further, this paper makes a comparison of the results obtained using the three phase four wire topology with per phase based dq control, and using natural reference frame based three phase control implemented by Plet *et al.* [13]. The purpose is to underscore the superior performance of the dq control in unbalanced conditions.

The remaining portion of this paper is organized in the following sequence. Section II discusses the sample system which is used in the simulation study. Implementation of per phase based dq control is discussed in Section III along with PWM generation unit. Section V presents concluding remarks, and section VI discusses future work.

II. SYSTEM DESCRIPTION

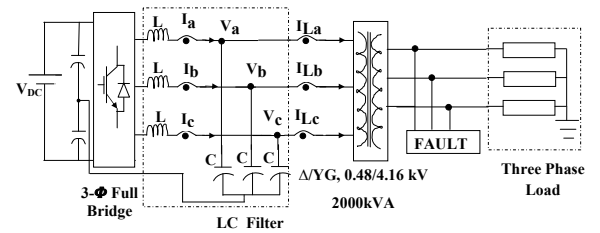


Fig. 1. Sample system for simulation.

The system under study consists of a 480V, 1760 kVA three phase four wire inverter supplied from a dc voltage source of 900V, as shown in Fig. 1. An adequate margin of dc voltage is provided so that the output voltage will not distort even when the inverter is required to supply only maximum reactive power load. An LC output filter with inductance of $100\mu\text{H}$ and capacitance of $2500\mu\text{F}$ is designed to filter out higher order voltage harmonics, as described in [5]. Moreover, total parasitic resistance of the filter inductor and power MOSFET switches is considered to be $1.63\text{ m}\Omega$ [5]. Unbalanced three phase load is supplied through an interface transformer (0.48/4.16 kV) connected at output terminals of the inverter. Due to the requirement to supply unbalanced loads, Δ/YG transformer is used. The load is modeled as an unbalanced constant impedance load (phase A=587 kVA, phase B=352 kVA, phase C=117 kVA) rated at 4.16 kV, and 0.8 lagging power factor. This model is chosen because an inverter is required to supply unbalanced loads with reactive power when it operates in stand alone mode.

III. INVERTER CONTROL IN STAND ALONE MODE

MATLAB/SIMULINK[®] based phase control module implemented in dq rotating reference frame is shown in Fig. 2. Each phase is controlled separately so that the inverter would have the flexibility to supply unbalanced loads, as well as supply loads through healthy phases even during faults. Phase to neutral voltages V_{abc} measured across filter capacitors,

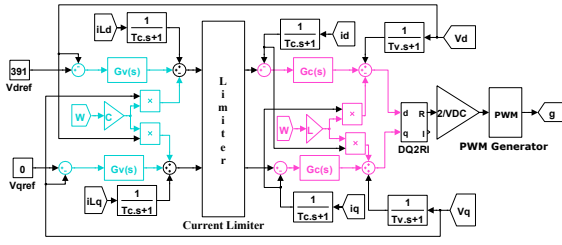


Fig. 2. Simulink based phase control module in dq frame.

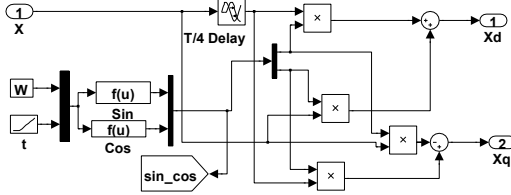


Fig. 3. Transformation of a phase signal into dq components.

phase currents through filter inductors I_{abc} , and load currents (inverter output currents) I_{Labc} are transformed to their dq equivalents through dq transformation module as shown in Fig. 3. Details of this phase dq transformation module is presented in [4], [20]. Output obtained through this transformation module is fed to the control module of Fig. 2. Input variable X in Fig. 3 represents any one of the phase currents or phase to neutral voltages, while the output variables X_d and X_q represent their dq equivalents.

Phase control module of Fig. 2 contains inner current control loop (magenta colored) and outer voltage control loop (cyan colored). Parameters of PI controllers in the current loop are chosen in such a way that they regulate the inductor currents fast enough to respond to load dynamics [5]. Similarly, parameters of the controllers in the voltage loop (outer loop) are chosen in such a way that voltage regulation

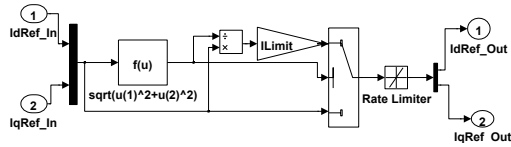


Fig. 4. Current limiter

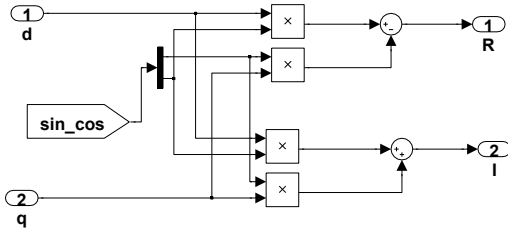


Fig. 5. Transformation of dq components into stationary frame (DQ2RI).

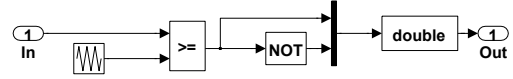


Fig. 6. PWM generator.

is fast enough without interference with current controllers [5]. Parameters of these controllers are manually tuned in such a way that the inverter yields good performance both in grid connected and standalone mode of operation. Current and voltage controllers used in this simulation study had transfer functions of $G_c = (0.5s + 8.158)/s$ and $G_v = (1.92s + 291)/s$ respectively. Moreover, feed forward loops of load currents and output voltages are implemented as shown in Fig. 2 to make the inverter's response fast during transients [5]. Delays in current feedback and feed forward paths are represented by a time constant of $T_c = 10 \mu s$. Similarly, delays in voltage feedback and voltage feed-forward paths are represented by a time constant of $T_v = 10 \mu s$.

The outer voltage control loop generates current references I_{dRef} and I_{qRef} which require to be regulated within thermal limit of the semiconductor switches in order to prevent the equipment damage during overloads and faults. Before reference currents are fed to the inner current control loop, I_{dRef} and I_{qRef} both are passed through current limit module as shown in Fig. 4. This current limiter limits phase currents through the filter inductor which ultimately limits current through semiconductor switches (MOSFETs). Outputs (d and q axes outputs) of current controller are transformed to their equivalent in phase domain by DQ2RI module shown in Fig. 5. Output signal R of the DQ2RI module is the modulating signal to be used for the generation of PWM [4]. This modulating signal is normalized by dividing it with $V_{DC}/2$. Sinusoidal pulse width modulation (SPWM) is implemented with this normalized modulating signal, as shown in Fig. 6.

IV. SIMULATION RESULTS

In the control module, inverter's current limit was set at $I_{Limit} = 3300$ A (peak), which is 110% of the inverter current rating. Inverter model constructed in MATLAB/SIMULINK[®] using the control modules described in section III was simulated with balanced and unbalanced loads at different power factor. Results presented in this paper correspond to highly unbalanced, constant impedance load (Phase A = 587 kVA, Phase B = 352 KVA, Phase C = 117 KVA) at 0.8 lagging power factor. The inverter was started with the load already connected, and it was able to pick up the load in a couple of cycles. Responses to four different types of shunt faults are recorded at both side of interfacing transformer while faults were imposed at the load side. Two sets of current waveforms I_{abc} and I_{Labc} are measured at the inverter side. Overcurrent protection of inverter is concerned with the inductor current (I_{abc}), while output current (I_{Labc}) represents inverter's response to the external circuitry connected to it. Measurements of inverter side quantities are used for control and protection of inverter,

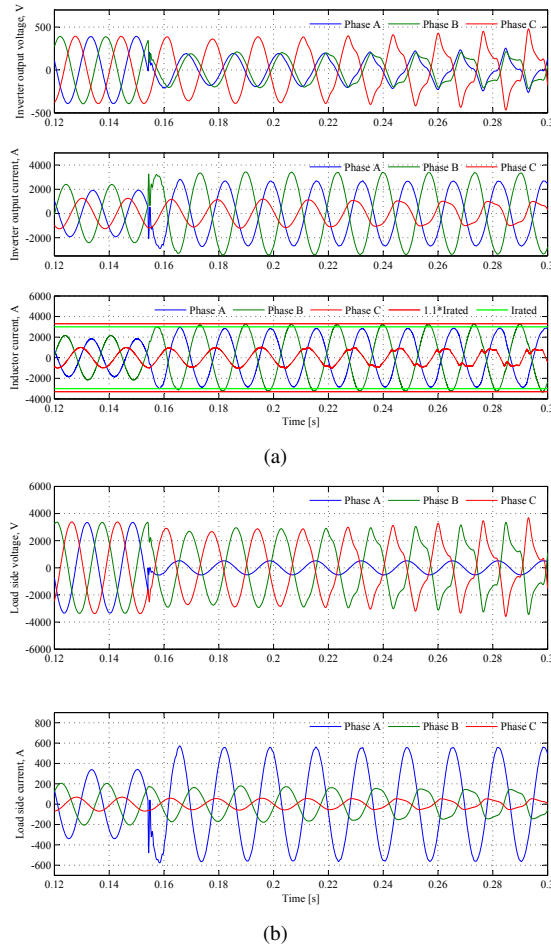


Fig. 7. Response of 3 Phase three wire inverter to single line to ground (AG) fault: (a) measurement at the inverter side of transformer, (b) measurement at the load side of transformer

while measurements of voltages and currents at the load side of the transformer are to be used for protection of microgrid. All faults were imposed at $t = 0.15416$ s (nine and quarter cycle after starting) once the inverter attained a steady state operation after starting. These faults had fault resistance of 1Ω .

In the first part of study, neutral point of filter capacitor was disconnected from the dc midpoint to change the three phase four wire inverter topology into three phase three wire. Then its responses to different types of faults were observed, out of which the response to single line to ground fault (AG) is presented in this paper. Fig. 7(a) shows voltage and current waveforms recorded at the inverter side of the interconnecting transformer, while Fig. 7(b) shows voltage and current waveforms recorded at the load side of the transformer. It can be seen that the three phase three wire inverter controlled in per phase dq coordinate has good performance during normal loads even if the loads are highly unbalanced. However, it shows poor performance during unbalanced faults. Highly distorted output voltages are observed even after few cycles of operation with unbalanced faults. It is observed that the

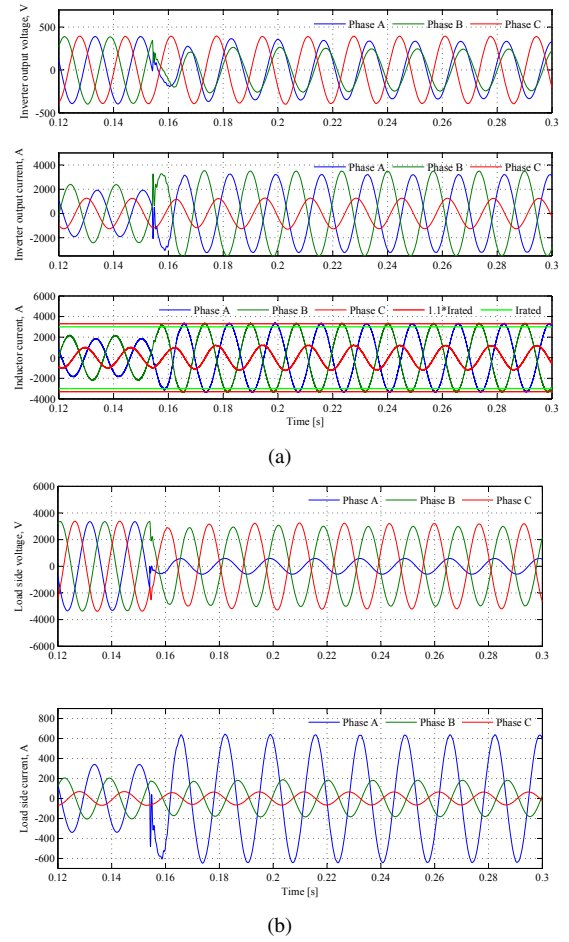


Fig. 8. Response of three phase four wire inverter to single line to ground (AG) fault: (a) measurement at the inverter side of transformer, (b) measurement at the load side of transformer

voltage waveforms of healthy phases are also severely affected (distorted). However, Inductor current, during the fault, is restricted within the limit of $3300A$ (110% of its rating) set for its thermal protection. Inverter's current rating and its thermal limit are shown by green and red horizontal lines respectively in the plot of inductor current vs time (Fig. 7(a)).

In the second part of the study, simulations were carried out with the three phase four wire inverter topology to observe its fault response. Fig. 8, 9, 10, and 11 show its responses to different types of shunt faults when the inverter was supplying the same prefault load as in case of the three phase three wire topology. Faults were imposed at the same location with the same fault resistance of 1Ω .

Fig. 8 shows the response of the inverter to a single line to ground fault imposed on phase A. It can be seen from Fig. 8 (a) that the inductor currents of phase A and phase B reach I_{Limit} , and remain within the limit. However, phase C current and phase C voltage remains unaltered at prefault values. Since single line to ground fault on phase A at the load side of a Δ/YG transformer appears as a line to line fault between phases A and B at the inverter side, it is obvious that phase C

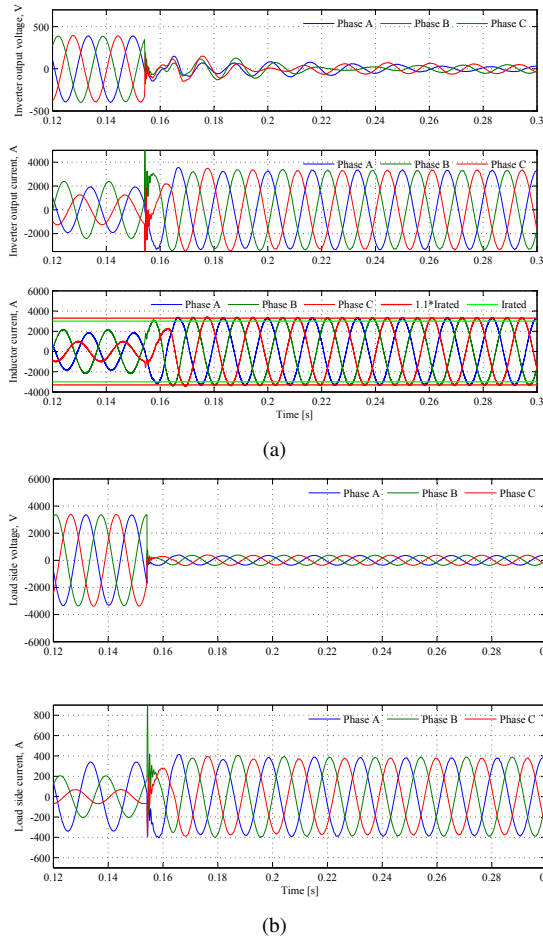


Fig. 9. Response of three phase four wire inverter to three phase fault: (a) measurement at the inverter side of transformer, (b) measurement at the load side of transformer

appears as the healthy phase. Fig. 8 (b) shows that the voltage and current waveforms at the load side of the inverter are not distorted.

Fig. 9 shows the response of the inverter to a balanced three phase fault. It can be seen that inductor currents and output currents are clean and harmonics free even during the fault. Moreover, the inductor currents are restricted at 110% of the inverter current rating. The inverter output voltage is unbalanced, and contains predominantly zero sequence component (notice the waveforms are in phase) for about six cycles as shown in Fig. 9 (a). However, the voltage waveforms measured at the load side of the transformer shown in Fig. 9 (b) are balanced and transient free immediately after the inception of fault. It is to be noted that microgrid protection will only deal with the output voltage and currents measured at the load side of transformer, where these quantities are perfectly balanced.

Fig. 10 shows the response of the inverter when a line to line fault between phases *B* and *C* was imposed. This line to line fault at the load side of the transformer is affecting the inductor currents and the output voltages in all three phases. It can be seen that the the inverter currents in the faulted

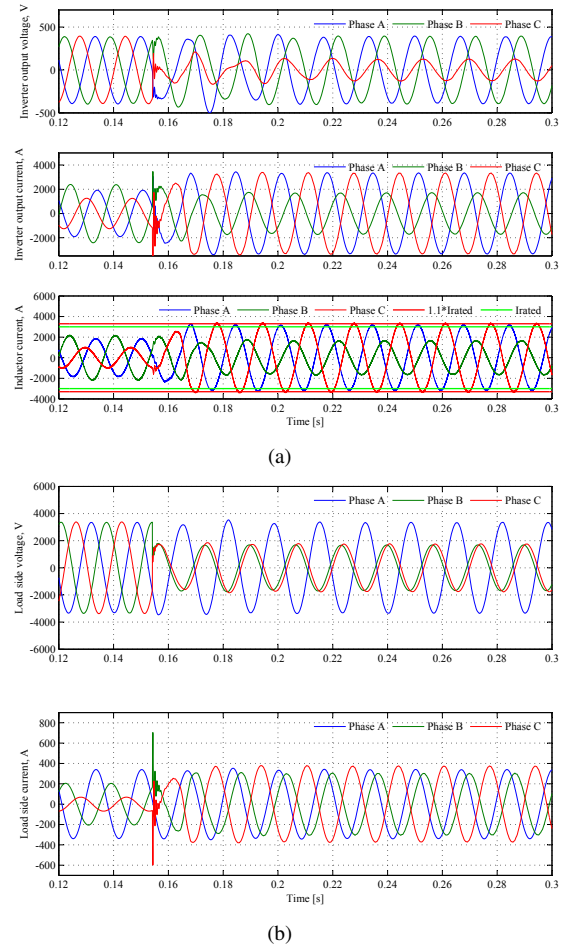


Fig. 10. Response of three phase four wire inverter to line to line (BC) fault: (a) measurement at the inverter side of transformer, (b) measurement at the load side of transformer

phases are restricted at I_{Limit} . Moreover, voltage and current waveforms recorded at the load side of the transformer are without distortion (harmonics free).

The response of the inverter to a double line to ground fault (BCG) is shown in Fig. 11. Inductor currents are well restricted within I_{Limit} as in the case of other faults. Also, output voltage and currents measured at the load side of the transformer are perfectly sinusoidal.

The results described here clearly illustrate two points:

- 1) The three phase four wire topology for inverters provide better performance during faults.
- 2) Considering the three phase four wire topology, the waveforms presented here using per phase based dq control are less distorted than the results reported by Plet *et al.* [13] using natural reference frame based three-phase control.

V. CONCLUSION

This paper describes the implementation of a three-phase inverter control in synchronously rotating frame, where each of the phases was separately controlled. A fault current limiter is also implemented. Three phase three wire and three phase four

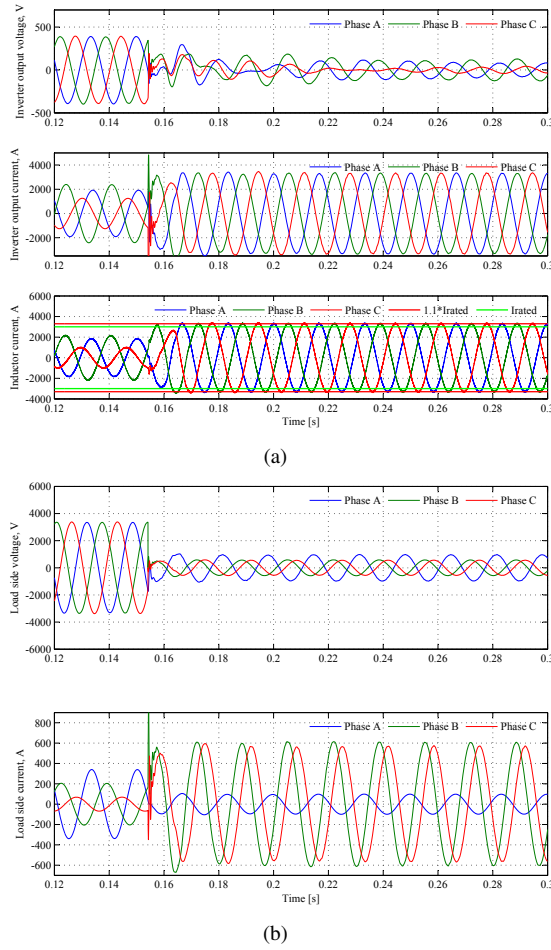


Fig. 11. Response of three phase four wire inverter to double line to ground (BCG) fault: (a) measurement at the inverter side of transformer, (b) measurement at the load side of transformer

wire inverter topologies are compared for their performance under fault. Simulation results show that three phase four wire topology gives better performance during unbalanced faults. For this topology, the results obtained using the proposed per phase based dq control yields better performance under faults compared with the results reported in literature using natural reference frame based control.

VI. FUTURE WORK

The inverter model described in this paper will be modified to work in the grid-connected mode, both with and without the presence of the substation source. A microgrid will be modeled with multiple inverters interfaced. Performance of such microgrid will be analyzed in grid-connected and islanded modes. Based on the results, appropriate fault models and protection system for such microgrids will be developed and tested.

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